Information Processing

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$

1. This question caries 40% of the marks.

a) For each of the following continuous-time signals, state whether they are causal or noncausal, periodic or nonperiodic, even or odd symmetric. If periodic, state the period.

i) $x_1(t) = |2\cos(3\pi t)|$ [2]

ii) $x(t) = \sin 2t + \sin 2\pi t$ [2]

b) Consider the signal

$$x(t) = \begin{cases} 1-t & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Describe, using illustrative sketches, the sequence of steps that must be taken to form $y(t)$ from $x(t)$.

i) $y(t) = x(-2t-1)$ [3]

ii) $y(t) = x(0.5t - 0.5)$ [3]

c) An LTI system $H(z)$ has impulse response $h[n] = 0.1^n u[n]$.

i) Derive and simplify an expression for the step response of $H(z)$, denoted $s[n]$. [3]

ii) Show how $h[n]$ can be obtained from your expression for $s[n]$. [3]

iii) Draw illustrative labelled sketches of $h[n]$ and $s[n]$. [2]

d) Consider a signal $x_1(t)$ with Fourier transform

$$X_1(j\omega) = \begin{cases} 2 & |\omega| < \omega_1 \\ 0 & \text{otherwise} \end{cases}$$

i) Find the signal $x_1(t)$ in terms of ω_1. Show your working. [3]

ii) Draw labelled sketches of $x_1(t)$ and $X_1(j\omega)$. [4]

iii) Now consider a signal $x_2(t)$ with Fourier transform

$$X_2(j\omega) = \begin{cases} 2 & |\omega| < \omega_2 \\ 0 & \text{otherwise} \end{cases}$$

and $\omega_2 = \pi\omega_1$.

Draw a labelled sketch of $x_2(t)$ and state the similarities and differences between $x_1(t)$ and $x_2(t)$. [2]

e) Consider the signal $x(t)$ shown in Fig. 1.1.

i) Write an expression for the signal $x(t)$. [1]

ii) Derive its Laplace transform $X(s)$ and show your working. [3]

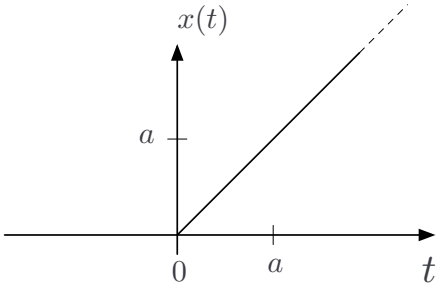


Figure 1.1

f) Consider the finite duration discrete-time signals $x_1[n]$ and $x_2[n]$. The nonzero values of these signals are given in Table 1. Find and sketch the signal $y[n] = x_1[n] * x_2[n]$ for the range $-5 \leq n \leq 20$ where $*$ indicates convolution. [4]

n	-3	-2	-1	0	1	2	3	4	5	6	7
$x_1[n]$		4	1	-3	-1						
$x_2[n]$							1	6	-2	3	

Table 1

g) Consider a discrete-time signal $x[n]$ with DTFT

$$X(e^{j\Omega}) = \begin{cases} e^{-\pi\Omega/2} & 0 \leq \Omega \leq \pi \\ e^{\pi\Omega/2} & -\pi < \Omega < 0. \end{cases}$$

Draw a sketch showing the magnitude spectrum of

$$X(e^{j\Omega}), \quad -2\pi \leq \Omega \leq 2\pi.$$

[2]

h) The discrete-time signal $x[n]$ with sample index n has z-transform $X(z)$ with *Region of Convergence* R_x. Showing each step of your working, prove that

$$x[-n] \overset{z}{\longleftrightarrow} X\left(\frac{1}{z}\right)$$

and state the *Region of Convergence* for this z-transform. [3]

2. This question caries 30% of the marks.

a) Summarize the similarities, differences and relationships between the Fourier transform and the bilateral Laplace transform in the context of continuous-time signals and systems. Include any relevant illustrative sketches. [7]

b) Consider the continuous-time LTI system $H(s)$ with input signal $x(t)$, output signal $y(t)$, and impulse response

$$h(t) = ae^{-4t} - be^{-3t}.$$

It is known that

$$y(t) = \frac{19}{20}e^{t} \quad \text{for } x(t) = e^{t}$$
$$y(t) = \frac{4}{5}e^{2t} \quad \text{for } x(t) = e^{2t}.$$

i) Write an expression for $H(s)$ and find the numerical values of the constants a and b. [5]

ii) Sketch the representation of $H(s)$ on the s-plane. [1]

iii) State the d.c. gain of this system. [1]

c) Using partial fraction expansion, find the inverse bilateral Laplace transform of the system described by

$$H(s) = \frac{s}{(s+3)(s^2+4s+5)} \qquad \text{ROC: } \mathrm{Re}(s) > -2.$$

[8]

d) A continuous-time LTI system is described by

$$\frac{d^2y(t)}{dt^2} - 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + x(t)$$

where $x(t)$ is the input signal and $y(t)$ is the output signal.

i) Find the corresponding system function, denoted $H(s)$, as the ratio of two polynomials. [3]

ii) Find the impulse response of this system, denoted $h_s(t)$, if the system is stable. [3]

iii) Find the impulse response of this system, denoted $h_c(t)$, if the system is causal. [2]

3. This question caries 30% of the marks.

a) Consider two signals

$$x_1[n] = 0.9^n u[n] \quad x_2[n] = 0.2^n u[n]$$

where $u[n]$ is the unit step function. Now consider another signal

$$y[n] = x_1[-n-2] * x_2[n-3]$$

where $*$ denotes convolution. Find the z-transform of $y[n]$. [6]

b) Consider a set of sound recordings of bird song. The recordings were made on a portable digital recording device with an analogue-to-digital converter operating at a fixed sampling frequency of 48 kHz with 16 bits per sample. The device has a built-in microphone with an associated microphone amplifier with variable gain.

The bits representing each sample are denoted $Q0$, $Q1$, ..., $Q15$. Bit $Q15$ is the sign bit so that $Q15 = 0$ for positive samples and $Q15 = 1$ for negative samples. The bits $Q0$, $Q1$, ..., $Q14$ represent the unsigned magnitude of the samples where $Q0$ denotes the Least Significant Bit and $Q14$ denotes the Most Significant Bit.

A study was made of the recordings. The bit values corresponding to the samples of the signals were examined separately for the periods of time when the birds were singing (denoted SONG) and for the periods of time when the birds were briefly silent between songs (denoted PAUSE). The study also included a spectral analysis of the recordings.

The conclusions of the study were as follows.

- Bits $Q13$ and $Q14$ are always zero in both SONG and PAUSE.
- Bits $Q0 \ldots Q4$ vary with uniform distribution in PAUSE.
- There is no significant energy above 11.5 kHz.

i) State the properties of the recordings that can be inferred from the study's conclusions. [3]

ii) State what signal processing could be advantageously applied to the recorded bird song signals and explain what advantages or disadvantages such processing would have. [2]

iii) State any recommendations you would give for improving the recording process without changing the recording device, and explain the nature of the expected benefits. [2]

c) A 60 Hz continuous-time cosinusoidal signal is sampled using a sampling frequency of 500 Hz. A total of 64 samples are acquired and stored as the discrete-time signal $x[n]$. A 64-point FFT is then applied to $x[n]$ to obtain the discrete spectrum $X[k]$ for $k = 0, 1, \ldots, 63$. Determine the value (or values) of k for which $X[k]$ has the greatest magnitude. [6]

d) Consider a discrete-time LTI system with impulse response $h[n]$ and system function $H(z)$. The pole-zero diagram for $H(z)$ is shown in Fig. 3.1.

i) Find expressions for $h[n]$ and $H(z)$. [2]

ii) Draw a block diagram of the implementation of the system function $H(z)$, denoting the input signal $x[n]$ and the output signal $y[n]$. [2]

iii) Draw a labelled sketch of the magnitude frequency response of $H(z)$ for the range of normalized angular frequency $0 \leq \Omega < \pi$.

[3]

iv) Determine the output, $y[n]$, when the input is

$$x[n] = 3.7 + 4.4\cos(n\pi/2) - 2\cos(n\pi/4).$$

[4]

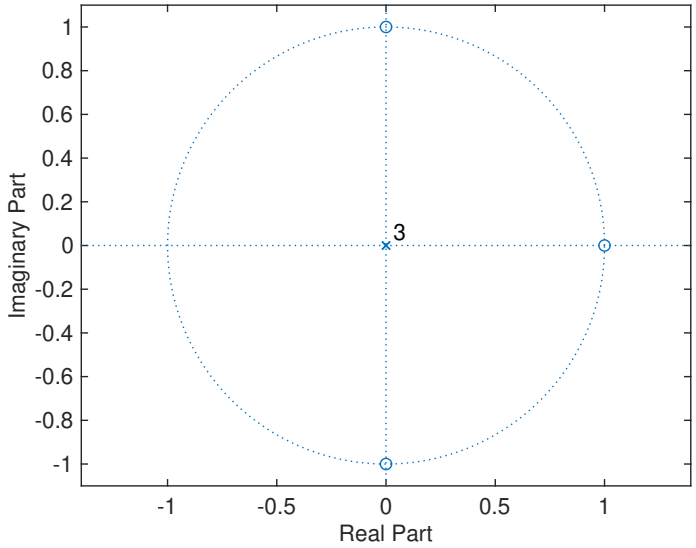


Figure 3.1